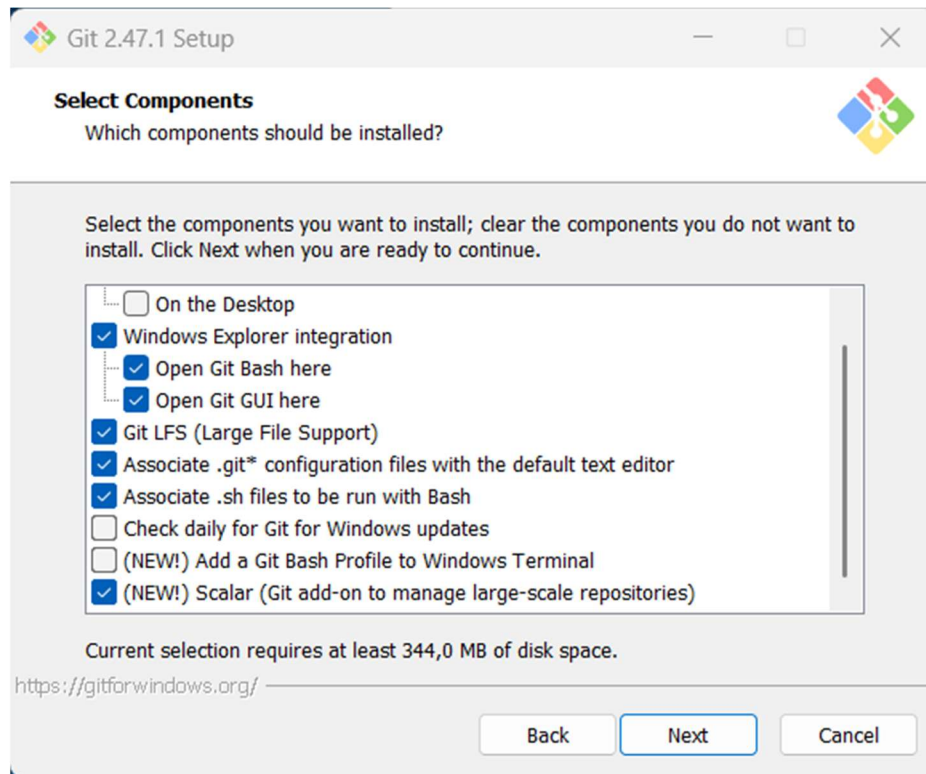
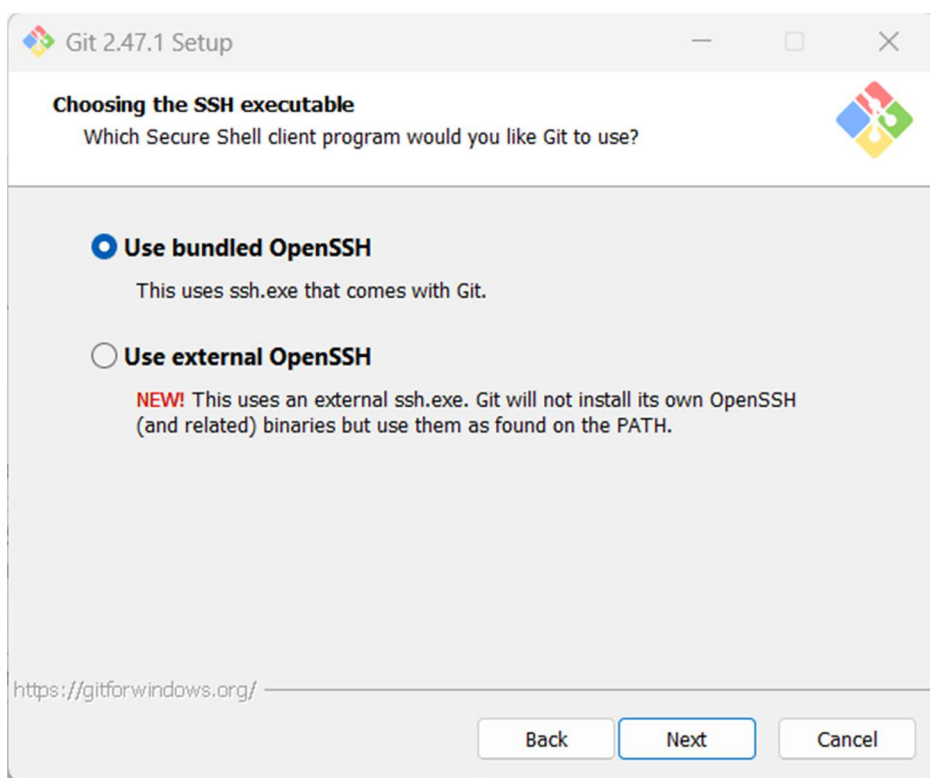
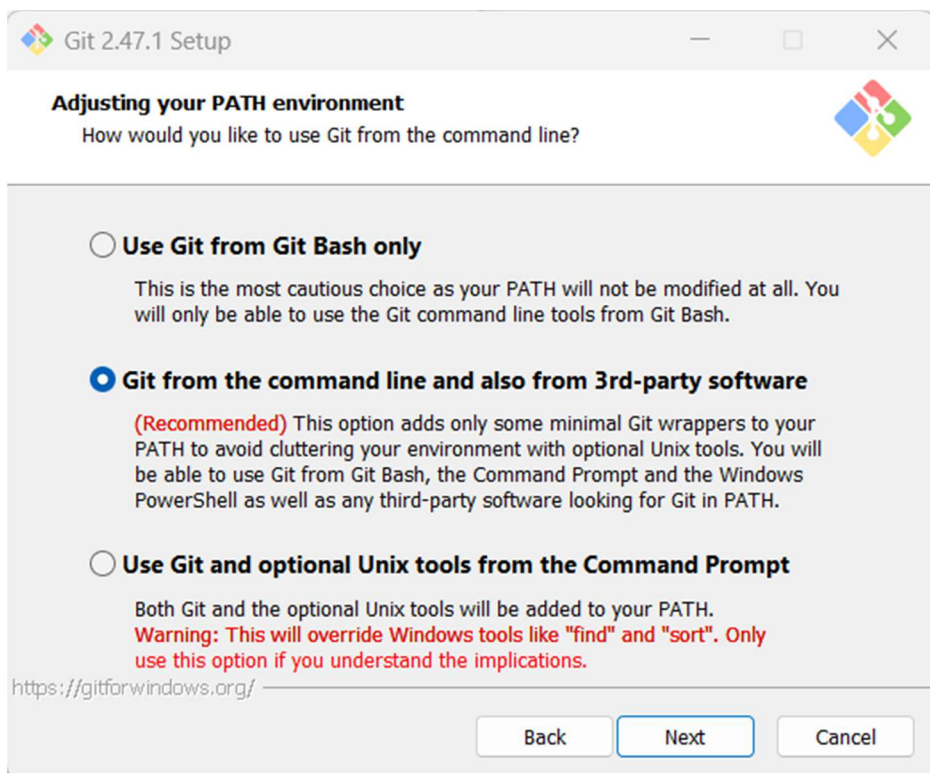


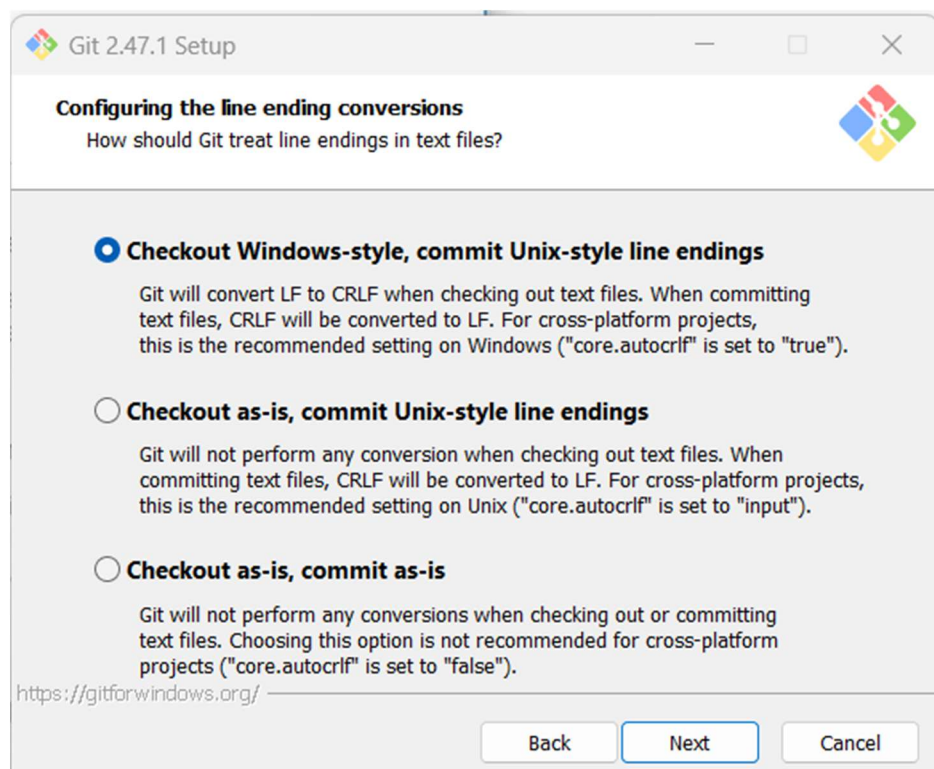
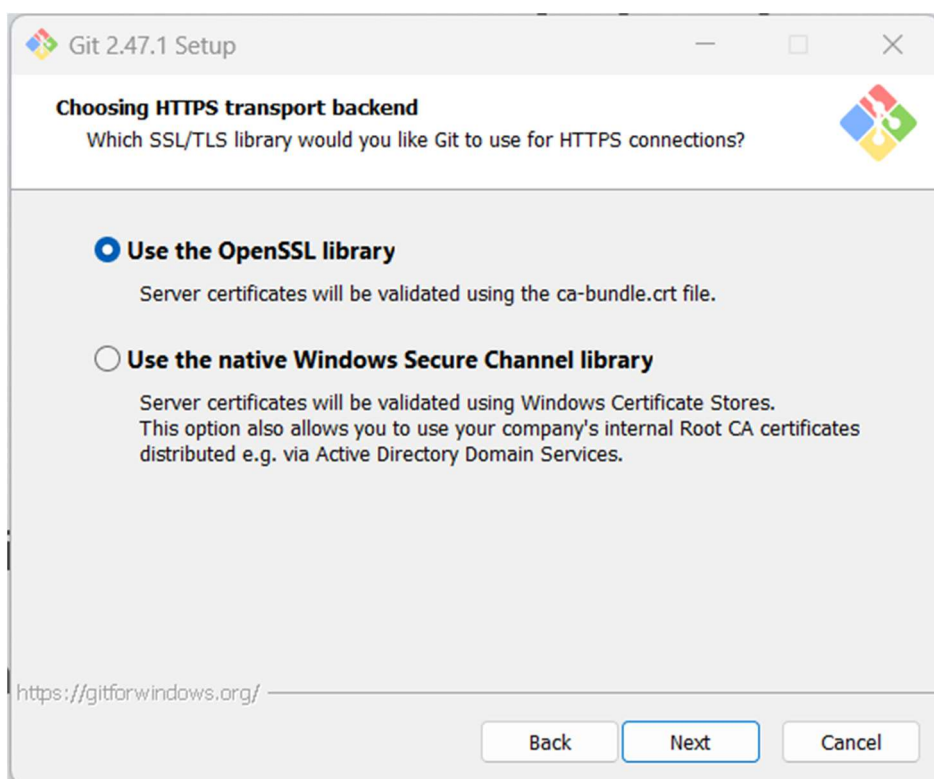
Atelier N° 9 : Master DSBD & IA

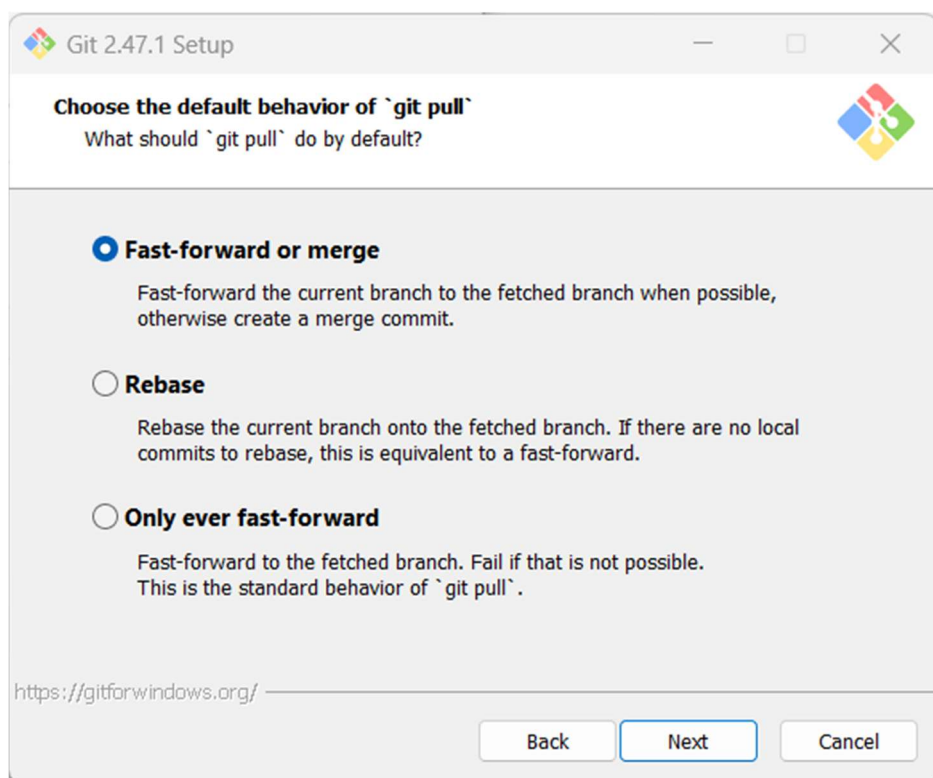
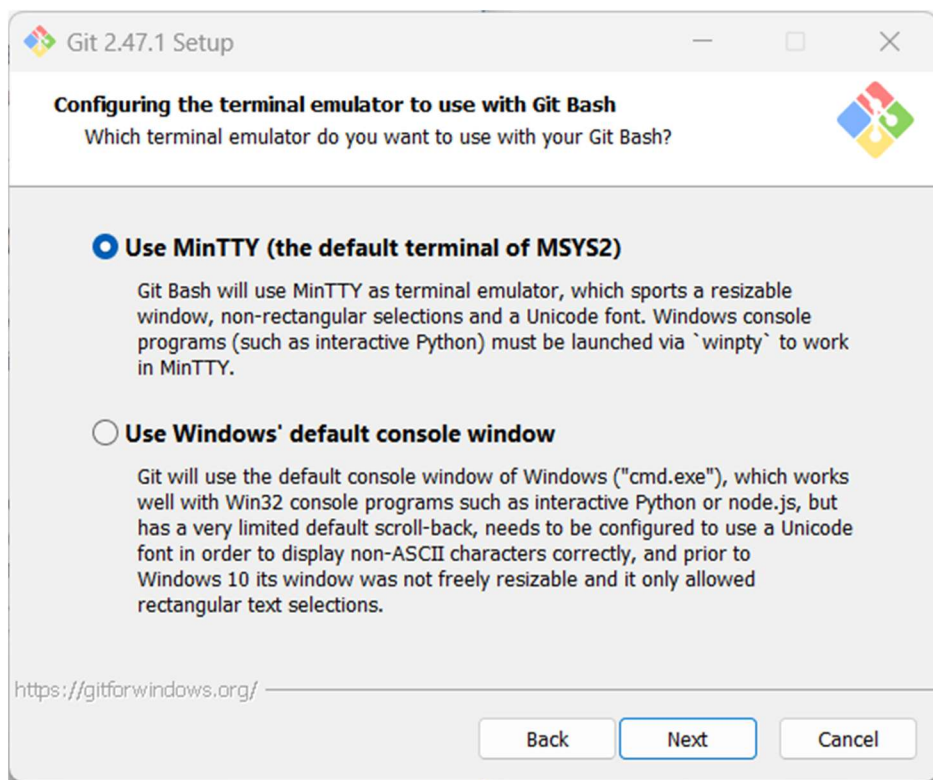
Système de contrôle de version Git

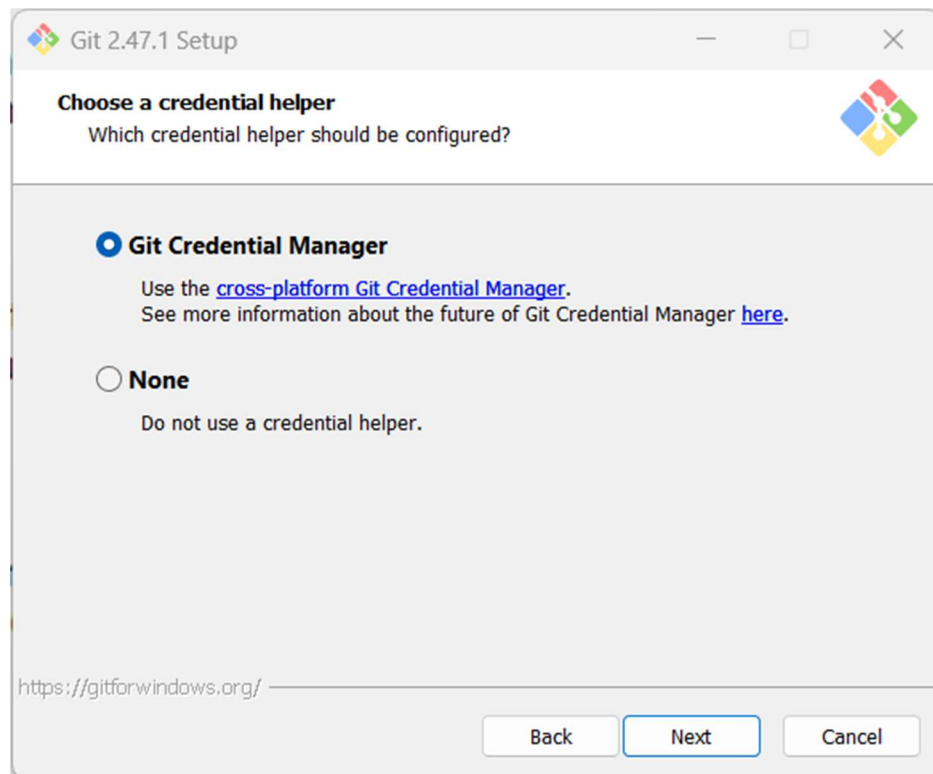
1. Installation







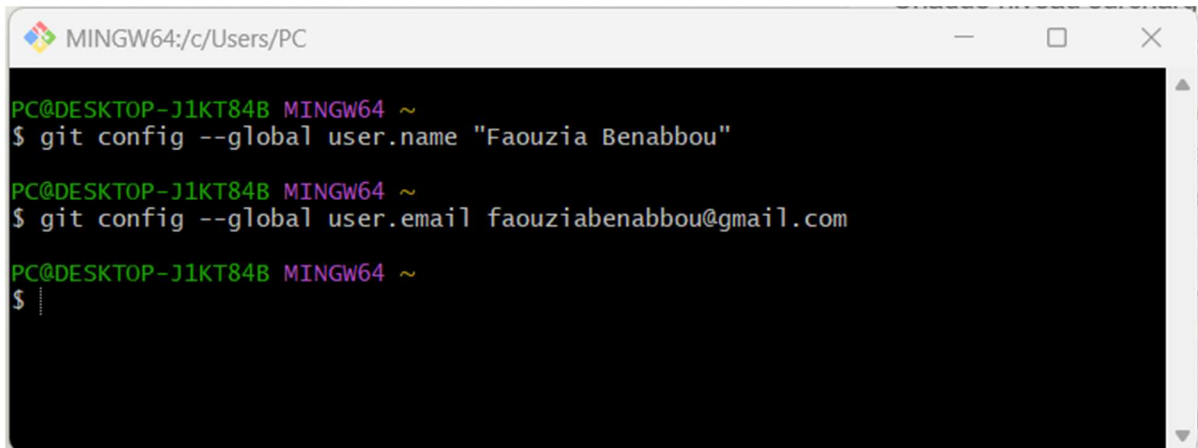




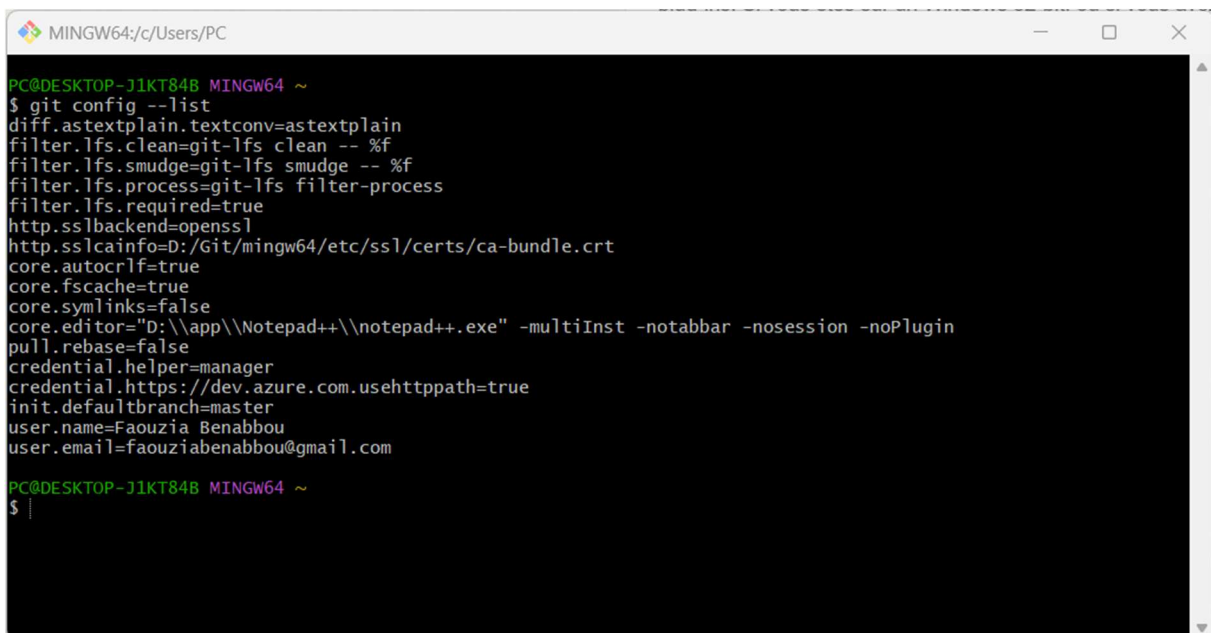
Vérifier :

The image shows a MINGW64 terminal window with the title bar 'MINGW64:/c/Users/PC'. The terminal output is as follows:
PC@DESKTOP-J1KT84B MINGW64 ~
\$ git version
git version 2.47.1.windows.1
PC@DESKTOP-J1KT84B MINGW64 ~
\$

2. Configurez l'identité

A terminal window titled 'MINGW64:/c/Users/PC' showing the execution of two git config commands. The first command sets the global user name to 'Faouzia Benabbou', and the second sets the global user email to 'faouziabenabbou@gmail.com'. The prompt is 'PC@DESKTOP-J1KT84B MINGW64 ~'.

Vérifier la liste des configurations :

A terminal window titled 'MINGW64:/c/Users/PC' showing the output of the 'git config --list' command. The output lists various configuration settings including diff, filter, http, core, credential, and user information. The prompt is 'PC@DESKTOP-J1KT84B MINGW64 ~'.

On peut par exemple activer les couleurs mais avant on va vérifier qu'elle est la configuration actuelle.

Pour revenir à la configuration par défaut des couleurs :

```
git config --global --unset color.ui
```

3. Démarrer un dépôt Git

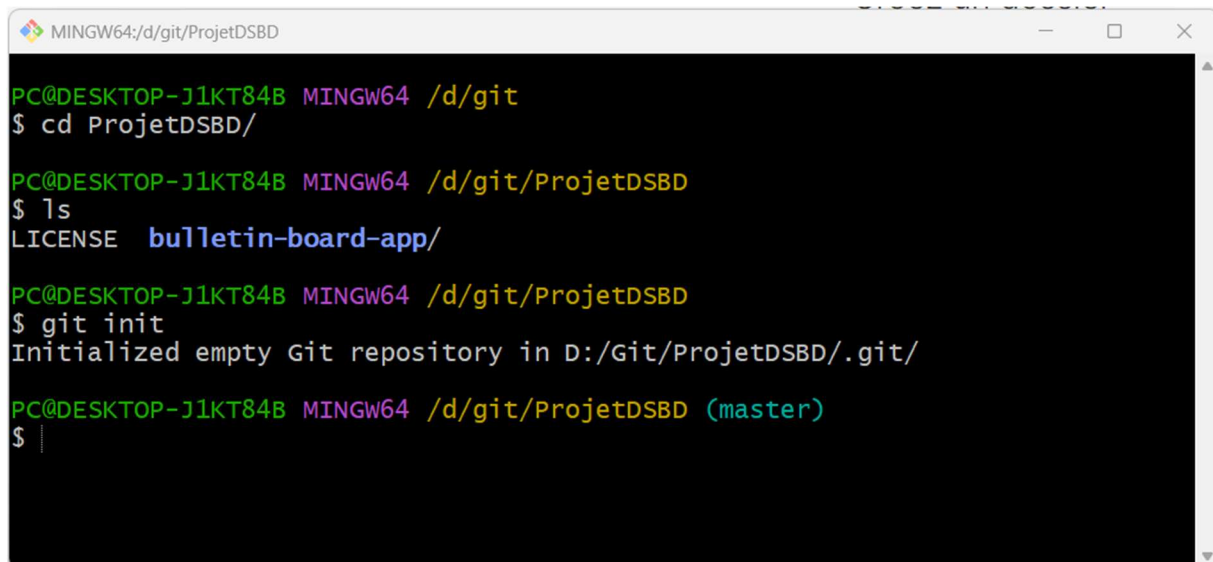
On peut démarrer un dépôt Git de deux manières.

- prendre un répertoire existant et le transformer en dépôt Git.

- cloner un dépôt Git existant sur un autre serveur.

Dans les deux cas on obtient un dépôt Git sur la machine locale, prêt pour y travailler ; nous allons utiliser la première méthode en utilisant le nom du répertoire où se trouve le projet.

Initialiser le dépôt projetDSBD :



```
MINGW64:/d/git/ProjetDSBD
PC@DESKTOP-J1KT84B MINGW64 /d/git
$ cd ProjetDSBD/

PC@DESKTOP-J1KT84B MINGW64 /d/git/ProjetDSBD
$ ls
LICENSE  bulletin-board-app/

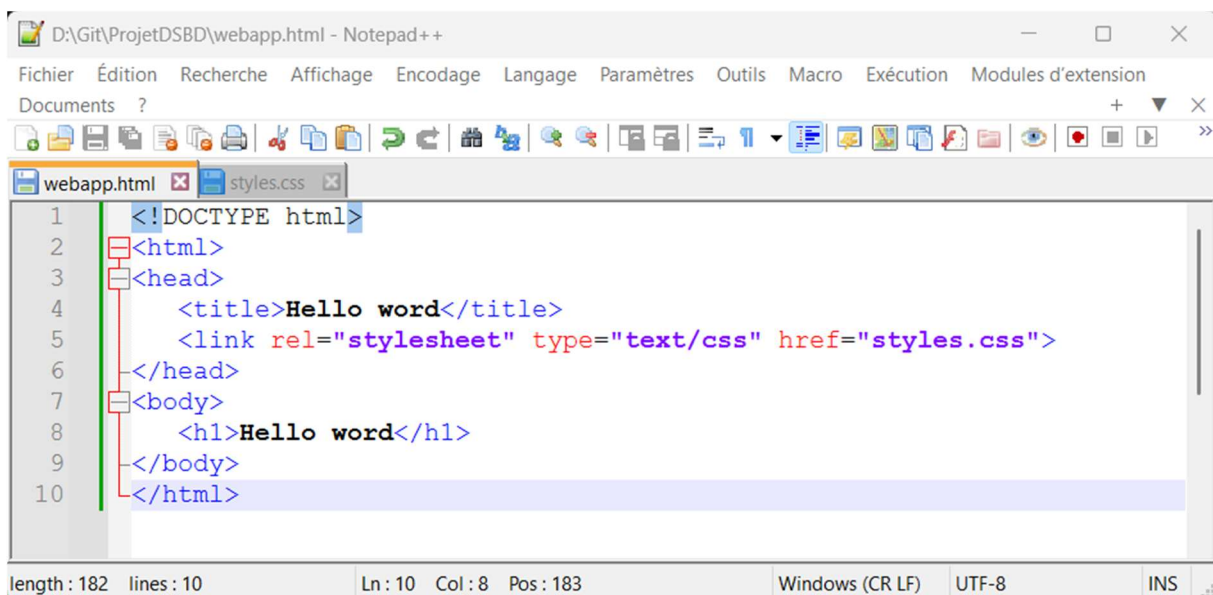
PC@DESKTOP-J1KT84B MINGW64 /d/git/ProjetDSBD
$ git init
Initialized empty Git repository in D:/Git/ProjetDSBD/.git/

PC@DESKTOP-J1KT84B MINGW64 /d/git/ProjetDSBD (master)
$
```

Un nouveau sous-répertoire nommé `.git` qui contient tous les fichiers nécessaires au dépôt : la configuration, les “logs”, les branches, etc. Pour l’instant, aucun fichier n’est encore versionné.

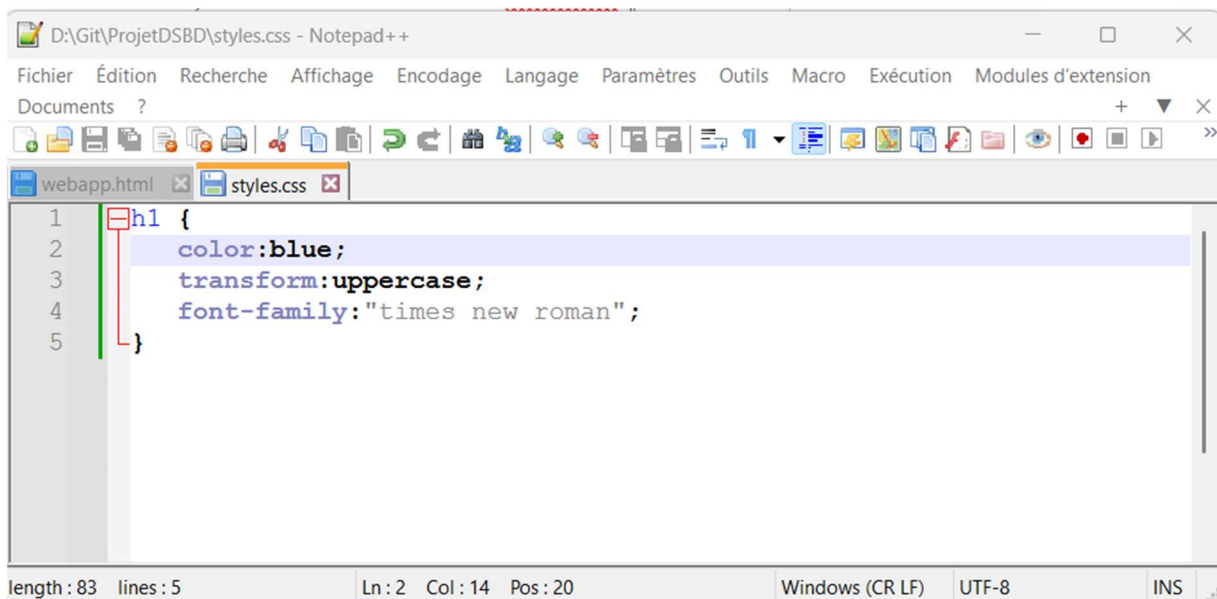
5. Création d’un projet

Une simple page web :



```
D:\Git\ProjetDSBD\webapp.html - Notepad++
Fichier  Édition  Recherche  Affichage  Encodage  Langage  Paramètres  Outils  Macro  Exécution  Modules d'extension
Documents ?
webapp.html  styles.css
1  <!DOCTYPE html>
2  <html>
3  <head>
4      <title>Hello word</title>
5      <link rel="stylesheet" type="text/css" href="styles.css">
6  </head>
7  <body>
8      <h1>Hello word</h1>
9  </body>
10 </html>
length: 182  lines: 10  Ln: 10  Col: 8  Pos: 183  Windows (CR LF)  UTF-8  INS
```

Avec les styles suivants :

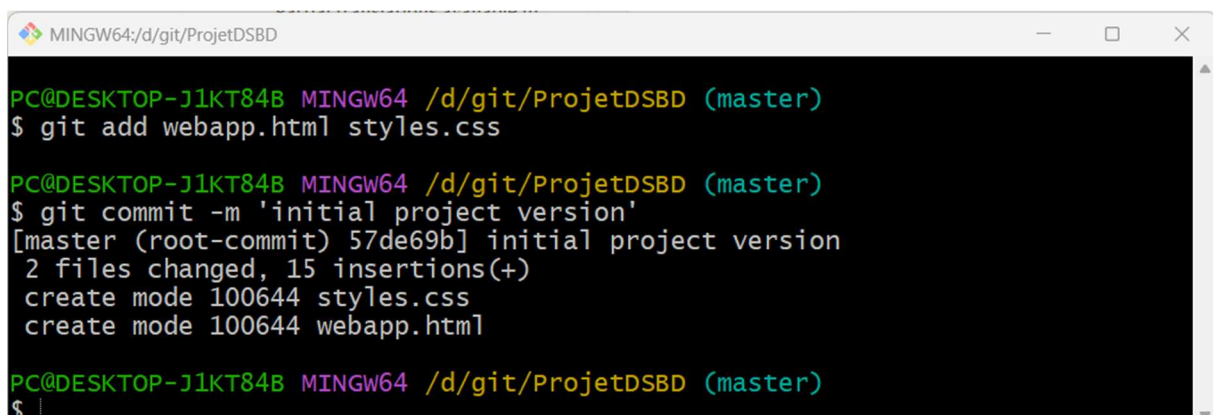


The screenshot shows a Notepad++ window with the file 'D:\Git\ProjetDSBD\styles.css' open. The menu bar includes 'Fichier', 'Édition', 'Recherche', 'Affichage', 'Encodage', 'Langage', 'Paramètres', 'Outils', 'Macro', 'Exécution', and 'Modules d'extension'. The toolbar contains various icons for file operations and editing. The editor shows the following CSS code:

```
1 h1 {  
2   color:blue;  
3   transform:uppercase;  
4   font-family:"times new roman";  
5 }
```

The status bar at the bottom indicates 'length: 83 lines: 5', 'Ln: 2 Col: 14 Pos: 20', 'Windows (CR LF)', 'UTF-8', and 'INS'.

Ces fichiers sont situés dans le dépôt Git (projetDSBD). Pour pouvoir créer une nouvelle version du projet, on doit indexer les deux fichiers par la commande « add ». Puis nous devons faire un « commit » où « initial project version » est un argument expliquant la raison de la modification.

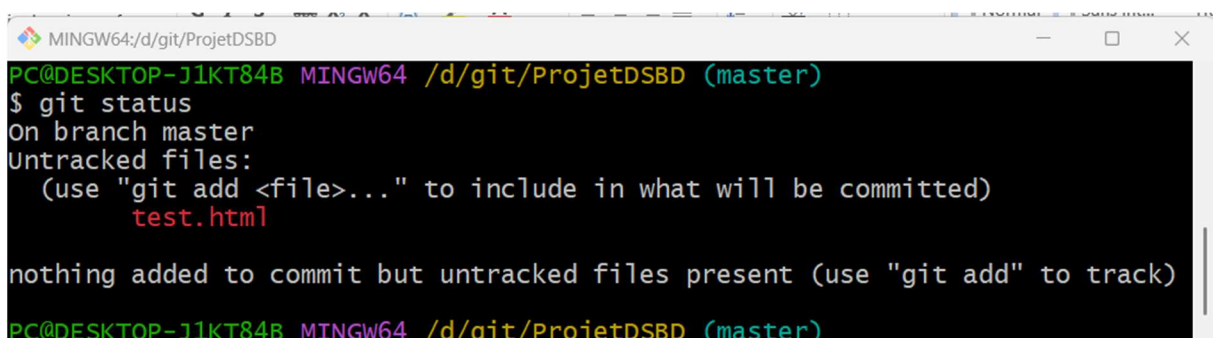


The screenshot shows a terminal window with the following commands and output:

```
PC@DESKTOP-J1KT84B MINGW64 /d/git/ProjetDSBD (master)  
$ git add webapp.html styles.css  
  
PC@DESKTOP-J1KT84B MINGW64 /d/git/ProjetDSBD (master)  
$ git commit -m 'initial project version'  
[master (root-commit) 57de69b] initial project version  
2 files changed, 15 insertions(+)  
create mode 100644 styles.css  
create mode 100644 webapp.html  
  
PC@DESKTOP-J1KT84B MINGW64 /d/git/ProjetDSBD (master)  
$
```

6. Suivre les modifications

Nous allons créer un autre fichier test.html sans lui appliquer « add » et « commit » (mettre ce que vous voulez), et nous allons voir si il est suivi dans git :



The screenshot shows a terminal window with the following commands and output:

```
PC@DESKTOP-J1KT84B MINGW64 /d/git/ProjetDSBD (master)  
$ git status  
On branch master  
Untracked files:  
  (use "git add <file>..." to include in what will be committed)  
    test.html  
  
nothing added to commit but untracked files present (use "git add" to track)  
  
PC@DESKTOP-J1KT84B MINGW64 /d/git/ProjetDSBD (master)
```

Indexer le fichier test.html et vérifier avec « git status ».

```
MINGW64/d/git/ProjetDSBD
PC@DESKTOP-J1KT84B MINGW64 /d/git/ProjetDSBD (master)
$ ls -l
total 3
-rw-r--r-- 1 PC 197121 83 Jan 1 14:24 styles.css
-rw-r--r-- 1 PC 197121 170 Jan 1 15:55 test.html
-rw-r--r-- 1 PC 197121 182 Jan 1 14:25 webapp.html

PC@DESKTOP-J1KT84B MINGW64 /d/git/ProjetDSBD (master)
$ git status
On branch master
Untracked files:
  (use "git add <file>..." to include in what will be committed)
       test.html

nothing added to commit but untracked files present (use "git add" to track)

PC@DESKTOP-J1KT84B MINGW64 /d/git/ProjetDSBD (master)
$ git add test.html

PC@DESKTOP-J1KT84B MINGW64 /d/git/ProjetDSBD (master)
$ git status
On branch master
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        new file:   test.html

PC@DESKTOP-J1KT84B MINGW64 /d/git/ProjetDSBD (master)
$
```

Annuler l'indexation du fichier test.html.

```
PC@DESKTOP-J1KT84B MINGW64 /d/git/ProjetDSBD (master)
$ git reset

PC@DESKTOP-J1KT84B MINGW64 /d/git/ProjetDSBD (master)
$ git status
On branch master
Untracked files:
  (use "git add <file>..." to include in what will be committed)
       test.html

nothing added to commit but untracked files present (use "git add" to track)

PC@DESKTOP-J1KT84B MINGW64 /d/git/ProjetDSBD (master)
$
```

7. Création d'une nouvelle branche

```
MINGW64:/d/git/ProjetDSBD

PC@DESKTOP-J1KT84B MINGW64 /d/git/ProjetDSBD (master)
$ git branch
* master

PC@DESKTOP-J1KT84B MINGW64 /d/git/ProjetDSBD (master)
$ git branch fonc1

PC@DESKTOP-J1KT84B MINGW64 /d/git/ProjetDSBD (master)
$ git branch
fonc1
* master

PC@DESKTOP-J1KT84B MINGW64 /d/git/ProjetDSBD (master)
$
```

Aller sur la branche fonc1 pour faire des modifications.

```
MINGW64:/d/git/ProjetDSBD

PC@DESKTOP-J1KT84B MINGW64 /d/git/ProjetDSBD (master)
$ git switch fonc1
Switched to branch 'fonc1'

PC@DESKTOP-J1KT84B MINGW64 /d/git/ProjetDSBD (fonc1)
$ ls
styles.css test.html webapp.html

PC@DESKTOP-J1KT84B MINGW64 /d/git/ProjetDSBD (fonc1)
$
```

Editer et modifier le fichier webapp et faire un commit.

```
MINGW64:/d/git/ProjetDSBD

PC@DESKTOP-J1KT84B MINGW64 /d/git/ProjetDSBD (fonc1)
$ git status
On branch fonc1
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
    modified:   webapp.html

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    test.html

no changes added to commit (use "git add" and/or "git commit -a")
PC@DESKTOP-J1KT84B MINGW64 /d/git/ProjetDSBD (fonc1)
$
```

```
PC@DESKTOP-J1KT84B MINGW64 /d/git/ProjetDSBD (fonc1)
$ git commit -a -m "version0.1 avec un tableau"
[fonc1 52ddfff] version0.1 avec un tableau
1 file changed, 7 insertions(+), 1 deletion(-)

PC@DESKTOP-J1KT84B MINGW64 /d/git/ProjetDSBD (fonc1)
```

Switcher sur le repo main et faites aussi des modifications sur le fichier webapp.html.

8. Fusion de deux versions dans le repo main et fonc1

On va essayer de fusionner les changements effectués dans fonc1 avec master.

```
MINGW64/d/git/ProjetDSBD
PC@DESKTOP-J1KT84B MINGW64 /d/git/ProjetDSBD (master)
$ git merge fonc1
Auto-merging webapp.html
CONFLICT (content): Merge conflict in webapp.html
Automatic merge failed; fix conflicts and then commit the result.

PC@DESKTOP-J1KT84B MINGW64 /d/git/ProjetDSBD (master|MERGING)
$ git status
On branch master
You have unmerged paths.
(fix conflicts and run "git commit")
(use "git merge --abort" to abort the merge)

Unmerged paths:
(use "git add <file>..." to mark resolution)
both modified:   webapp.html

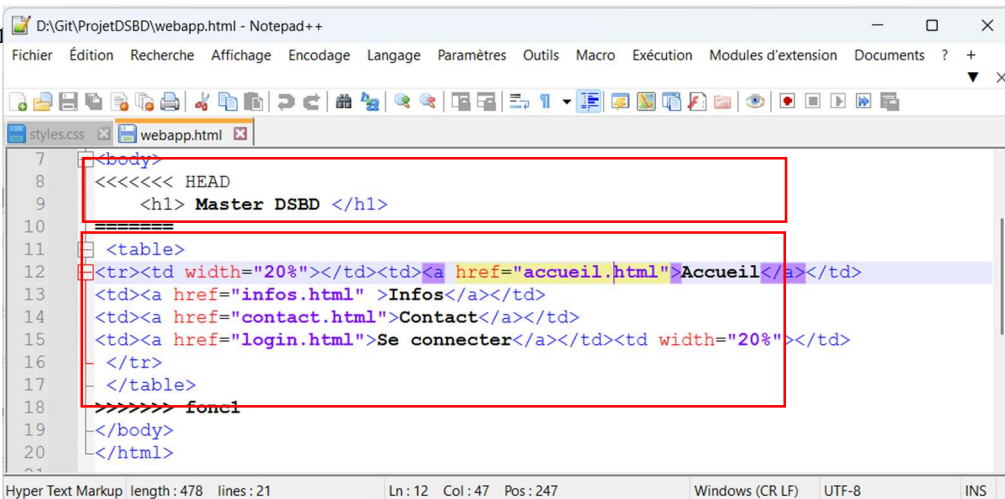
Untracked files:
(use "git add <file>..." to include in what will be committed)
.webapp.html.swp
test.html

no changes added to commit (use "git add" and/or "git commit -a")

PC@DESKTOP-J1KT84B MINGW64 /d/git/ProjetDSBD (master|MERGING)
$
```

On remarque qu'il y a un conflit, car les deux versions sont modifiées. Allez dans le fichier,

ici



```
D:\Git\ProjetDSBD\webapp.html - Notepad++
Fichier  Édition  Recherche  Affichage  Encodage  Language  Paramètres  Outils  Macro  Exécution  Modules d'extension  Documents  ?  +
webapp.html
7  <body>
8  <table>
9  <tr><td width="20%"></td><td><a href="accueil.html">Accueil</a></td>
10 </tr>
11 <tr><td width="20%"></td><td><a href="infos.html">Infos</a></td>
12 </tr>
13 <tr><td width="20%"></td><td><a href="contact.html">Contact</a></td>
14 </tr>
15 <tr><td width="20%"></td><td><a href="login.html">Se connecter</a></td></tr>
16 </table>
17 </body>
18 </html>
19
20
```

Dans la positionner dans la branche principale et modifier le fichier webapp pour résoudre le conflit et enregistrer. Puis exécuter les commandes d'indexation et commit :

```
MINGW64/d/git/ProjetDSBD
PC@DESKTOP-J1KT84B MINGW64 /d/git/ProjetDSBD (master|MERGING)
$ git add webapp.html
```

```
PC@DESKTOP-J1KT84B MINGW64 /d/git/ProjetDSBD (master|MERGING)
$ git commit -m "fusion de la version titre et tableau"
[master 98c7a30] fusion de la version titre et tableau
```

9. Inspection et Log

Exécuter les différentes commandes et analyser le rendu/

```
$ git log
$ git log --oneline --graph --decorate -all
$ git log -p
$ git log --stat --summary
$ git log --oneline
$ git log -n 5
git log --author="votre nom"
git log --pretty=format:"%h - %an, %ar : %s"
git log --since="2023-01-01" --until="2024-12-31"
git log --since="2024-01-01" --until="2025-01-03"
```

10. Travailler avec un repo à distance

Si vous avez déjà un projet sur github vous pouvez l'utiliser ou vous pouvez utiliser un repo public.

Clôner le projet :

```
MINGW64:/d/git/ProjetDSBD
PC@DESKTOP-J1KT84B MINGW64 /d/git/ProjetDSBD (master)
$ git clone https://github.com/faouziabenabbou/Syst-me-expert-Diagnostic.git
Cloning into 'Syst-me-expert-Diagnostic'...
remote: Enumerating objects: 13, done.
remote: Counting objects: 100% (13/13), done.
remote: Compressing objects: 100% (13/13), done.
remote: Total 13 (delta 3), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (13/13), 16.75 KiB | 816.00 KiB/s, done.
Resolving deltas: 100% (3/3), done.

PC@DESKTOP-J1KT84B MINGW64 /d/git/ProjetDSBD (master)
$ ls
Syst-me-expert-Diagnostic/  styles.css  test.html  webapp.html
```

```
MINGW64:/d/git/ProjetDSBD/Syst-me-expert-Diagnostic
PC@DESKTOP-J1KT84B MINGW64 /d/git/ProjetDSBD/Syst-me-expert-Diagnostic (main)
$ vim mycin.pl

PC@DESKTOP-J1KT84B MINGW64 /d/git/ProjetDSBD/Syst-me-expert-Diagnostic (main)
$ git status
On branch main
Your branch is up to date with 'origin/main'.

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   mycin.pl

no changes added to commit (use "git add" and/or "git commit -a")

PC@DESKTOP-J1KT84B MINGW64 /d/git/ProjetDSBD/Syst-me-expert-Diagnostic (main)
$
```

```
PC@DESKTOP-J1KT84B MINGW64 /d/git/ProjetDSBD/Syst-me-expert-Diagnostic (main)
$ git commit -a -m "Amélioration de l'interaction"
[main 1e19620] Amélioration de l'interaction
1 file changed, 1 insertion(+), 1 deletion(-)

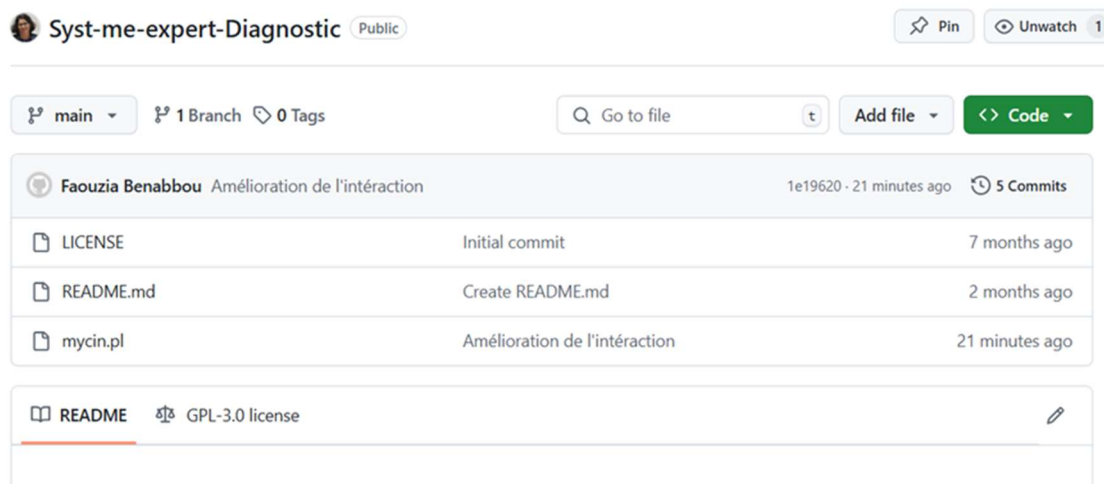
PC@DESKTOP-J1KT84B MINGW64 /d/git/ProjetDSBD/Syst-me-expert-Diagnostic (main)
$
```

```
MINGW64:/d/git/ProjetDSBD/Syst-me-expert-Diagnostic
PC@DESKTOP-J1KT84B MINGW64 /d/git/ProjetDSBD/Syst-me-expert-Diagnostic (main)
$ git config --global http.sslVerify false

PC@DESKTOP-J1KT84B MINGW64 /d/git/ProjetDSBD/Syst-me-expert-Diagnostic (main)
$ git push origin main
warning: ----- SECURITY WARNING -----
warning: | TLS certificate verification has been disabled! |
warning: -----
warning: HTTPS connections may not be secure. See https://aka.ms/gcm/tlsverify for more info
rmation.
info: please complete authentication in your browser...
warning: ----- SECURITY WARNING -----
warning: | TLS certificate verification has been disabled! |
warning: -----
warning: HTTPS connections may not be secure. See https://aka.ms/gcm/tlsverify for more info
rmation.
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 8 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 320 bytes | 320.00 KiB/s, done.
Total 3 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To https://github.com/faouziabenabbou/Syst-me-expert-Diagnostic.git
979cc22..1e19620 main -> main

PC@DESKTOP-J1KT84B MINGW64 /d/git/ProjetDSBD/Syst-me-expert-Diagnostic (main)
$
```

On vérifie sur GitHub.



Syst-me-expert-Diagnostic Public

main 1 Branch 0 Tags

Go to file Add file Code

Commit	Message	Time
1e19620	Amélioration de l'interaction	21 minutes ago
979cc22	Amélioration de l'interaction	21 minutes ago
...	...	...
...	...	...
...	...	...

README GPL-3.0 license

Télécharger le projet (si une autre personne l'a modifié)

```
MINGW64:/d/git/ProjetDSBD/Syst-me-expert-Diagnostic
PC@DESKTOP-J1KT84B MINGW64 /d/git/ProjetDSBD/Syst-me-expert-Diagnostic (main)
$ git config --global http.sslVerify true

PC@DESKTOP-J1KT84B MINGW64 /d/git/ProjetDSBD/Syst-me-expert-Diagnostic (main)
$ git push origin main
Everything up-to-date

PC@DESKTOP-J1KT84B MINGW64 /d/git/ProjetDSBD/Syst-me-expert-Diagnostic (main)
$ git pull origin main
From https://github.com/faouziabenabbou/Syst-me-expert-Diagnostic
* branch          main          -> FETCH_HEAD
Already up to date.

PC@DESKTOP-J1KT84B MINGW64 /d/git/ProjetDSBD/Syst-me-expert-Diagnostic (main)
$
```